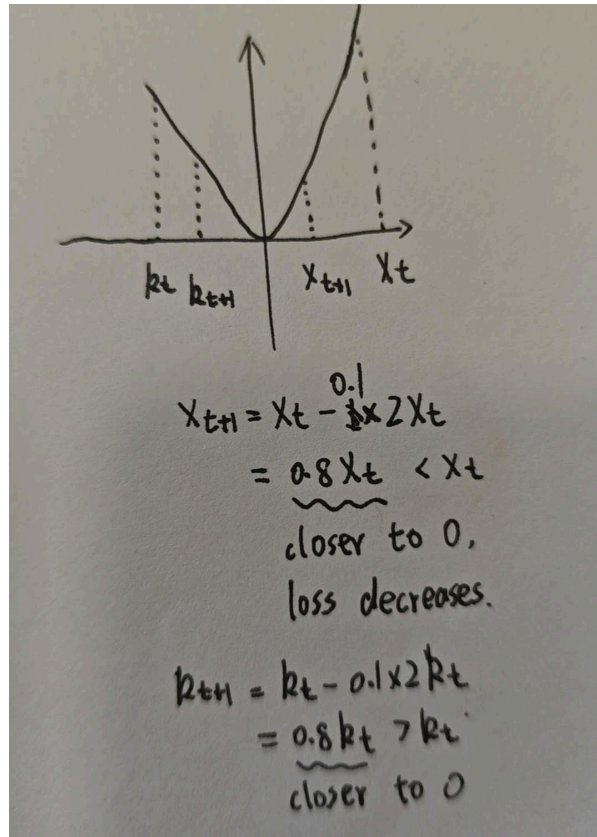


A justification for Backpropagation

When I first met the formula of BP algorithm, I had no reason why it looks like this:

$$x_{t+1} = x_t - \alpha \frac{\partial L}{\partial x_t}$$

, of course until to the time when I'm writing, I still had no reason. The only method for me to understand this it is: It works well in $f(x) = x^2$:



In recent days, I was learning gradient-based MAB algorithm and I tried to figure out the derivation. During asking AI, I found I felt uncomfortable about the formula of backpropagation, so I asked AI to give me the mindset behind the writing of the formula, and here is the mindset that AI told me that will be displayed with an example:

Let's say we have a loss function $f(x) = x^2$, then:

$$f(x + \Delta x) = f(x) + \frac{\partial f(x)}{\partial x} \Delta x$$

We know our goal is to make $f(x)$ small as much as possible, so for $\Delta x \rightarrow 0$, if

$$\Delta x = -\alpha \frac{\partial f(x)}{\partial x}, \alpha \rightarrow 0^+, \text{ then:}$$

$$\begin{aligned}
f(x + \Delta x) &= f(x) + \frac{\partial f(x)}{\partial x} \Delta x \\
&= f(x) - \alpha \left(\frac{\partial f(x)}{\partial x} \right)^2 \\
&\leq f(x)
\end{aligned}$$

, which means if $\Delta x = -\alpha \frac{\partial f(x)}{\partial x}$, then our loss function will not increase. We can also extend the derivation from $2D$ to ND space:
set:

$$\begin{aligned}
X &\in \mathbb{R}^{N \times 1} \\
\frac{\partial f(X)}{\partial X} &\in \mathbb{R}^{N \times 1} \\
f(X) &\in \mathbb{R}
\end{aligned}$$

then:

$$\begin{aligned}
f(X + \Delta X) &= f(X) + \left(\frac{\partial f(X)}{\partial X} \right)^T \Delta X \\
\Delta X &= -\alpha \frac{\partial f(X)}{\partial X} \\
f(X + \Delta X) &= f(X) - \alpha \left\| \frac{\partial f(X)}{\partial X} \right\|^2 \\
&\leq f(X)
\end{aligned}$$

so:

$$\begin{aligned}
x_{t+1} - x_t &= -\alpha \frac{\partial f(x)}{\partial x} \\
x_{t+1} &= x_t - \alpha \frac{\partial f(x)}{\partial x}
\end{aligned}$$

In conclusion, if $x_{t+1} = x_t - \alpha \frac{\partial f(x)}{\partial x}$, then the target function will decrease, but there is a limitation: $\alpha \rightarrow 0$, which is impractical in real situation, so I think it could be viewed as a theoretical justification.